LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 1: Meet Scratch: Sprites, the Stage and Your First Script

Module 1 | 45 minutes | Algorithms and sequence | Grades 3–5 · ages 8–10

Objective: Find the stage, sprites, block palette and code area, and build a script that runs blocks in order (a sequence).

Success criteria

- I can point to the stage, a sprite, the block palette and the code area.
- I can build a script that starts with the green flag.
- I can explain why the order of blocks matters.

Key vocabulary

- **algorithm**: A list of steps to do a task, in order.
- **sequence**: Instructions that run one after another, in order.
- **sprite**: A character or object in a Scratch project.
- **stage**: The area where a Scratch project plays.
- **script**: A stack of blocks snapped together.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Robot Teacher game: students instruct you to walk to the desk; discuss why exact, ordered instructions matter.
5 min	**Watch**	Play the lesson video (captions on). Pause after the screen tour and ask students to point to the stage and code area.
12 min	**Build along**	Students open Scratch and follow the six build-along steps. Circulate; check every student has a green-flag script that runs.
12 min	**Perform**	Robot puzzles 1–2, matching and ordering activities in Kitabash. Encourage “Run” before “Check”.
6 min	**Probe**	Zig-zag stairs puzzle and the lemonade algorithm. Fast finishers start the worksheet challenge.

5 min	**Wrap-up**	Exit ticket. Two students show their script; class predicts what it does before it runs.

Unplugged warm-up

Robot Teacher (5 min): you are the robot. Students give you instructions to walk from the door to your desk. Follow them exactly — walk into the wall if they forget to say “turn”. Ask: why did the robot fail?

Teaching the idea

A program is a sequence of instructions that a computer follows exactly, in order. In Scratch, instructions are blocks that snap into scripts; a script runs from top to bottom when its hat block’s event happens. Sprites act on the stage.

Analogy: A script is like a recipe: the same steps in a different order can give a very different result.

Live demo: Project Scratch on the board. Build “move 10 / say Hello!” live, then swap the blocks and ask students to predict the difference before you click.

Questions to ask

Ask	Listen for
What will the sprite do first?	The block at the top of the script.
What happens if we swap these two blocks?	The actions happen in the other order and the sprite may end somewhere else.
Why didn’t the robot (me) stop at the desk?	The instructions didn’t say to stop/turn — computers do exactly what we write.

Differentiation

Support
Pair students. Give the printed blocks card from the worksheet so they can point to each block before finding it on screen. Let them use one “move (3) steps” block in puzzle 1.

Stretch
Challenge: make the sprite walk to each corner of the stage and say a different message at each corner.

Common misconceptions

Misconception	How to fix it

The computer knows what I mean	Computers do exactly what you write, in order — nothing more, nothing less.
All the blocks in a script happen at the same time	Inside one script, blocks run one after another from top to bottom.
Clicking a block in the palette adds it to my script	Drag the block into the code area and snap it under another block.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

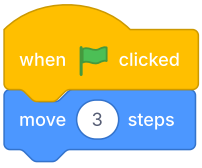
Exit ticket
Draw a 3-block script and write what the sprite will do, in order.

Homework
Write an algorithm (5 steps) for a morning routine. Swap two steps and write what would go wrong.

Answer key — Kitabash activities

1. First steps (perform · block code)

One solution:

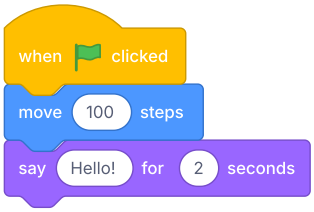


2. Code Lab: your first script (perform · code lab)

Goals checked: Starts with the green flag · Pip moves · Pip says something

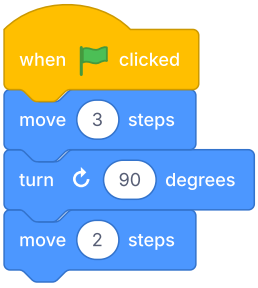
One solution:

Pip



3. Around the corner (perform · block code)

One solution:



4. Match each part of Scratch to what it does (perform · matching)

Stage → Where your project plays · Sprite → A character that follows your code · Block palette → Where you find all the blocks · Code area → Where you snap blocks into scripts

5. Read the script (perform · multiple choice)

Answer: It says “Hi!” for 2 seconds

6. Zig-zag stairs (probe · block code)

One solution:

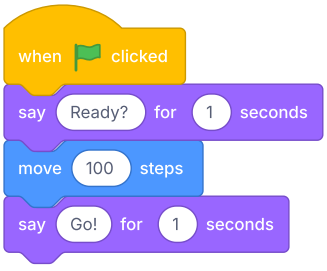


7. Code Lab: put it in order (probe · code lab)

Goals checked: Says “Ready?” first · Walks between the two messages

One solution:

Pip



8. Explain your thinking (probe · free response)

Look for: first, then, order. **Example:** First squeeze the lemon into a glass. Then add water. Then add sugar. Last, stir it. If I stirred before adding the sugar, the sugar would not mix in, so the order matters.

Answer key — worksheet

1. **Name the parts:** Stage · Sprite · Block palette · Code area
2. **Follow the code:** Up 2 squares, turn right, right 3 squares: Pip stops in row 2 (from the top), column 4 (from the left).

3. **Lemonade in order:** 4 Stir and drink · 1 Squeeze the lemon · 3 Add sugar · 2 Pour in water (water and sugar may swap).
4. **Your own algorithm:** Any clear, ordered steps, e.g. First walk 3 steps forward. Then turn left. Next walk 5 steps. Last sit down.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.